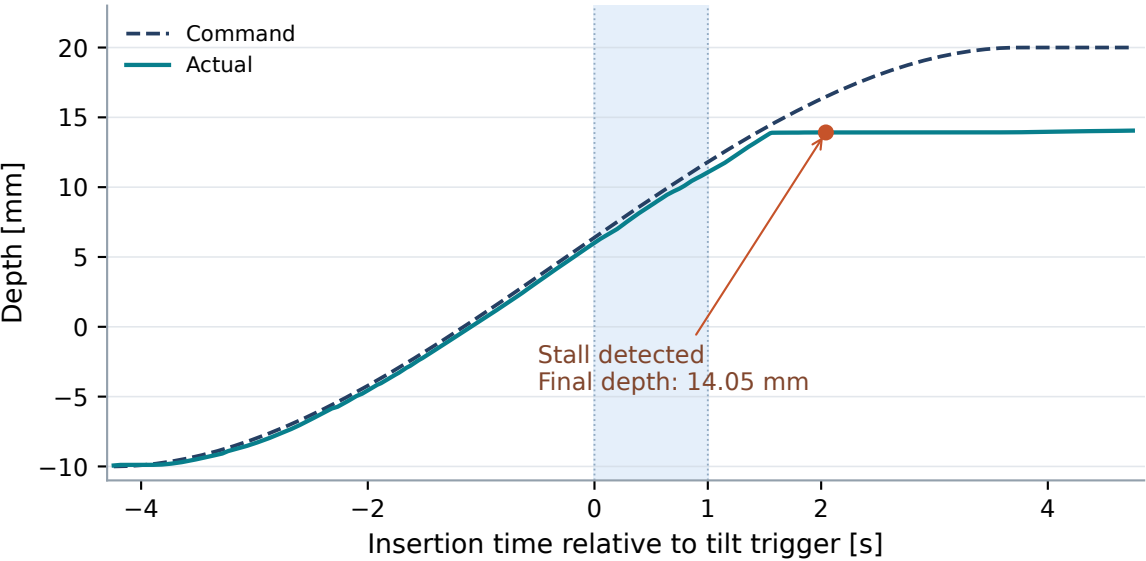


An inserted stall resists straight withdrawal, then releases

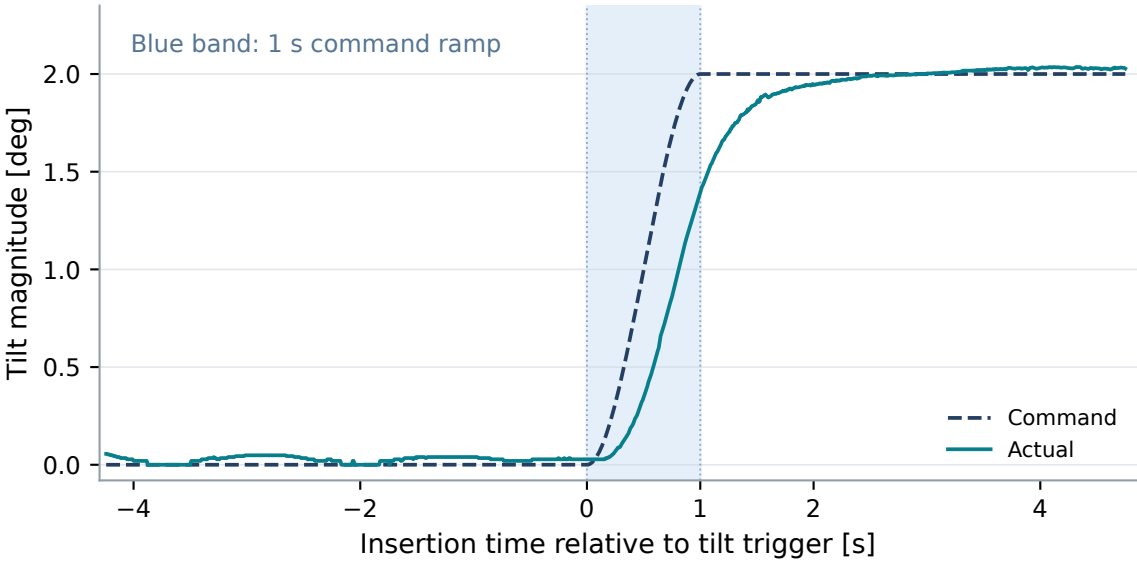
0.2 mm radial clearance · 2° pitch starts at 6 mm actual depth · original continuous withdrawal

Increasing pull initially produces very little peg motion. Force then drops as withdrawal accelerates; clearance succeeds.

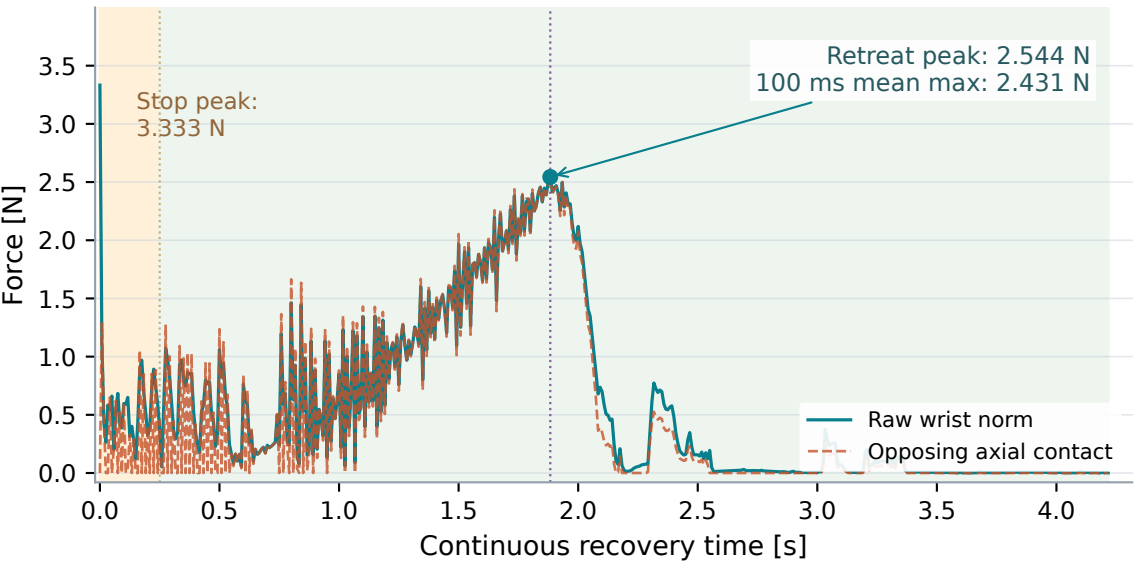
Insertion stops short of the 20 mm target



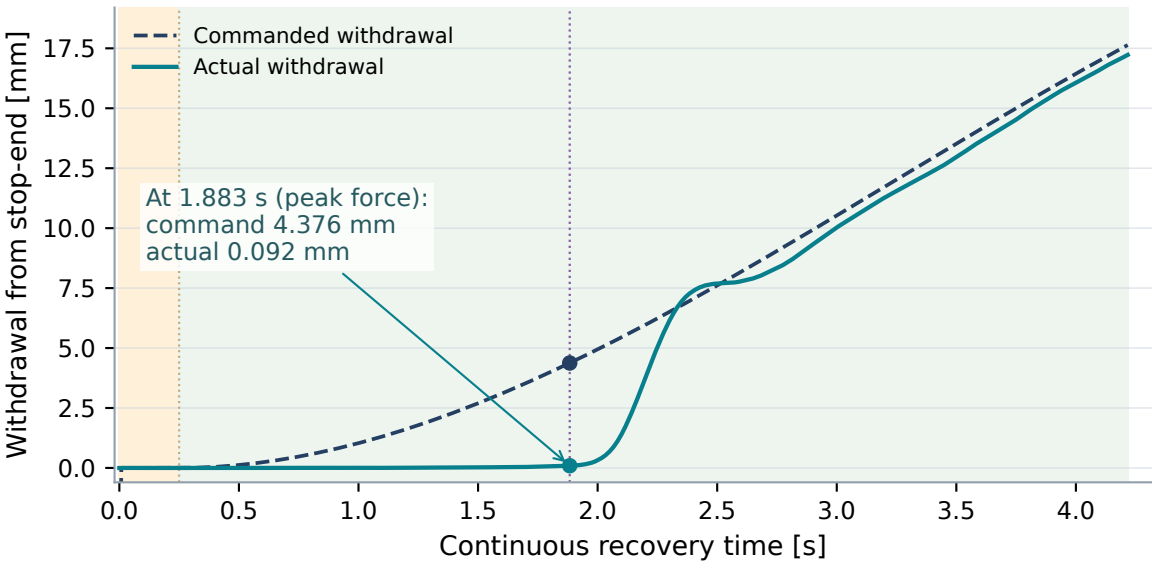
The time-driven tilt ramp completes despite the stall



Retreat load rises while motion remains small



The peg releases and then follows the retreat



Amber: 0.25 s stop. Green: withdrawal. Dotted purple: retreat force peak. Clearance confirmed at 4.217 s.

Wrist norm is an uncalibrated reaction signal; opposing contact = $\max(0, -\text{world contact } F_z)$. The higher stop peak is separate from the retreat peak.

This candidate passes numerical screening, but physical validity and timestep convergence still need confirmation.